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# Combining sensitivity analysis and scaling analysis for nuclear thermal-hydraulic studies

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ABSTRACT

The scaling analysis of an accidental nuclear transient can identify the possible scale distortions between scaled down facilities and a real power plant. Advanced scaling approaches exist today in order to identify and quantify these distortions. There is also significant ongoing progress in evaluating the impact of distortions on a thermal–hydraulic transient using simulations of an Integral Effect Test (IET) and of a reference reactor transient using mature system codes. This paper presents some parallel application of a scaling analysis and of the General Sensitivity Analysis (GSA), a statistical tool used to identify the impact of a series of pre-selected input parameters e.g. physical models, to a specific Figure of Merit (FoM). This work identifies common and complementary potentials between the scaling analysis and the GSA. A counterpart test between BETHSY and LSTF facilities for a Pressurized Water Reactor (PWR) Small Break LOCA (SBLOCA) is analyzed in order to show the benefits of this strategy. The results suggest some guidelines to promote the combined use of GSA and scaling analysis.

## 1. Introduction

The scaling analysis (U.S. NRC, 2005; IRSN, 2017) ; (OECD/NEA, 2017) consists of a series of activities to investigate the impact of the design choices of an Integral Effect Test (IET) performed in a reduced scale Integral Test Facility (ITF) to simulate a reactor transient. When a Phenomena Identification and Ranking Table (PIRT) is established, the transient is split into successive time phases also called Phenomenological Windows (PhWs) characterized by some dominant phenomena. The scaled facility should reproduce as least these relevant phenomena to guarantee the good representativeness at the reactor scale. Then, the scaling analysis includes the following steps: (i) the selection of Figures of Merit (FoM) and Parameters of Interest (PoIs) that characterize the general behavior of PhWs, (ii) the similarity analysis of dominant and influent processes using balance equations for the PoIs written for the reactor transient and for the IET and (iii) the evaluation of the order of magnitude of the''scaling equations'' terms for both reactor and experiment, which represent involved physical processes in order to (iv) define the design criteria for scaled down ITF and Boundary and Initial Conditions (BIC) for the IET that minimize distortions at least for the most important processes. Some design scaling approaches (Nahavandi et al., 1979) exist that formalized these processes e.g. the linear scaling (OECD/NEA, 2017), the Power-To-Volume Scaling (PTVS) (Carbiener and Cudnik, 1969), or the Power-To-Mass Scaling (PTMS) (Liu et al., 1997; Liu et al., 1998) . Despite the important effort to reduce scale distortions, biases remain unavoidable. The complexity of thermal–hydraulic systems and the presence of several constraints (design, costs…) bring to inevitable distortions in experimental facilities. For example, ITFs that were designed using the PTVS method cannot respect the scaling length in all three directions, which result in distortions of 3D phenomena. Again, in horizontal pipes (hot leg, cold leg and intermediate leg) distortions exist in PTVS. This is because although the Froude number is preserved to respect stratification, the onset of droplet entrainment is distorted. Bestion et al. (Bestion, 2024, Bestion et al., 2024) listed some common and inevitable distortions in nuclear systems. Some advanced scaling methods as the Three-Level Scaling (3LS) (D'Auria et al., 2012, Ishii et al., 1998), the Hierarchical Two-Tiered Scaling (H2TS) (Jr et al., 2015), the Fractional Scaling Analysis (FSA) (Catton et al., 2009; Novozhilov, 1997; Zuber, 1991) or the latest Dynamical System Scaling (DSS) (Reyes, 2015; Reyes and Frepoli, 2015; Bestion and D'Auria, 2022) propose a well-structured method to design the facilities and analyze, identify and classify distortions between facilities and the reactor scale in a-posteriori scaling analysis. These

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**Nomenclature**

| | |
|---|---|
| H | Volume averaged specific enthalpy (J $kg^{-1}$) |
| h | Local specific enthalpy (J $kg^{-1}$) |
| $\dot{M}$ | Mass flow rate (kg $s^{-1}$) |
| P | Pressure in a control volume (Pa) |
| t | Time (s) |
| V | Control volume ($m^3$) |
| $W$ | Power (W) |
| $\nu'_{k,p}$ | Partial derivative of specific volume of phase k with respect to pressure ($m^4$ $s^2$ $kg^{-2}$) |
| $\nu'_{k,h}$ | Partial derivative of specific volume of phase k with respect to enthalpy (m $s^2$ $kg^{-1}$) |
| $\rho$ | Density (kg $m^{-3}$) |

*Subscript*

| | |
|---|---|
| l | Liquid |
| v | Vapor |
| grav | gravity |
| tot | total |
| in | Inlet |
| out | outlet |
| acc | accumulator |
| br | break |
| i | interphase |
| core | core |
| k | Phase k |

*Acronyms*

| | |
|---|---|
| BETHSY | Boucle d'Etude ThermoHydraulique SYst'eme |
| CATHARE | Code Avancé de THermohydraulique pour les Accidents de Réacteurs à Eau |
| CEA | Commissariat à l'Energie Atomique et aux énergies alternatives |
| EDF | Electricité de France |
| FoM | Figure of Merit |
| FRC | Fractional Rate of Change |
| FSA | Fractional Scaling Analysis |
| GSA | General Sensitivity Analysis |
| H2TS | Hierarchical Two-Tiered Scaling |
| HPIS | High Pressure Injection System |
| LSC | Loop Seal Clearing |
| LSP | Loop Seal Plugging |
| PWR | Pressurized Water Reactor |
| PIRT | Phenomena Identification and Ranking Table |
| PCT | Peak Clad Temperature |
| PIRT | Phenomena Identification and Ranking Table |
| PoI | Parameter of interest |
| SB-LOCA | Small-Break Loss Of Coolant Accident |
| SG | Steam Generator |
| IET | Integral Effect Test |
| ITF | Integral Test Facility |

methods were initially applied without support of numerical simulations until system codes reached a sufficient high level of maturity. When system codes have been extensively validated and proved to be able to predict SETs and IETs with a reasonable accuracy, they can be used to provide more precise evaluation of the process terms of scaling equations. Some works have shown as the advanced scaling methods may be in support to more complex methodologies allowing to structure Phenomena Identification and Ranking Tables (PIRTs) (Bestion, 2024, Bestion et al., 2024) or identifying distortions coming from (i) the geometrical facility design (Joon Yoon et al., 2022; Gretton et al., 2005) (ii) the code meshing or (iii) the initial conditions of a chosen IET (D'Auria et al., 2012). Nevertheless, some scaling issues remain and further analyses are necessary. This work proposes some reflections and some guidelines in order to solve some of these unsolved issues:

- Zuber et al. (Catton et al., 2009) proposed some acceptability criteria for distortions that remain rather subjective. Criteria do not depend on the impact of distortions on FoM or Parameter of Interest (PoI) such as a primary mass inventory, a primary pressure, a core flow-rate, etc. They fix some boundaries (e.g. a factor 2) to the acceptable ratio of non-dimensional numbers relative to an IET and to the reference reactor. These non-dimensional numbers are the Effect Metrics for the FSA or the Pi-Groups for the H2TS. CEA is working on a more detailed, reliable and objective way, to define the impact and acceptability of distortions (Zuber et al., 2007) and some additional reflections are presented in this paper;
- The scaling analysis performed at system scale allows identifying in which component of the loop the distortion is expected (Ciechocki et al., 2022). Then a following analysis may be conducted at the component scale in order to identify the process which impacts the particular PoI. An example is presented in (Carnevali, 2024) for a Small Break LOCA (SB-LOCA) transient and a comparison between LSTF and BETHSY facilities and the PWR case. The scaling analysis applied to the primary system of the facilities and the PWR identified and quantified distortions between the facilities and the reference reactor. These distortions affected the FoM e.g. the Peak of Cladding Temperature (PCT). This study showed that the asymmetrical loop seal clearing in the facilities were not well predicted by the CATHARE system code. In such case, the scaling analysis must be performed without using codes but by simple order of magnitude analysis and using measurements in the IET. Accurately predicting Loop Seal Clearing (LSC), especially with asymmetry between loops, remains a challenge even for mature codes due to its high sensitivity to subtle inter-loop variations. Pressure equations were written for the primary system's upstream and downstream regions of the intermediate leg. The goal was to identify the process that controls the duration between Loop Seal Plugging (LSP) and Clearing (LSC) (Carnevali, 2024) and to analyze how it affects core uncovery and the subsequent temperature excursion. Such a-posteriori scaling analysis performed when experimental results are available may be used to investigate the possible distortions of LSP & LSC between reactor and scaled experiments. This is the top-down approach useful to identify where the distorted phenomena may occur. The CEA is working to a *generalized distortion correction strategy* (Zuber et al., 2007; Ciechocki et al., 2022) in order to identify, quantify and correct, if possible, distortions through an analysis of their potential impact on the transient. A systematic use of hybrid calculations that corrects the scaling distortions in IET calculations one by one will provide a clear understanding of what explains the differences between IET results and the code simulation results of a reactor transient. Also, these hybrid calculations will provide new more efficient acceptability criteria for scaling distortions since they show the effect of each distortion on the Figure of Merit itself. This analysis at macroscopic system and component scales should be complemented by a local approach focusing on the physical processes that are modelled in code by specific closure laws. This last step is useful for a more detailed investigation of the origin of the distortion (design/experimental-distortion, meshing/simulation-distortion…) and to draw the attention on the scalability of the most sensitive models and on requirements for their validation matrix;
- Scaling methods such as the H2TS or the FSA applied with simple order of magnitude evaluation of process terms is useful to

understand what controls the change of a PhW but do not provide temporal and inter-processes dependencies among the different PhWs.

In 2018 the Commissariat à l'Energie Atomique et aux Energies Alternatives (CEA) (D'Auria et al., 2012; Carnevali, 2024) ; (Bestion, 2024) ; (Martinez-Quiroga and Reventos, 2014) , France, started an important work focused on the scaling issues with the support of CATHARE code in the context of the BEPU methodology (Ciechocki et al., 2024). A key research area currently being studied is the complementarity between scaling and sensitivity analysis. This paper presents that research and proposes guidelines for future works. The scaling analysis presented hereafter was already conducted at CEA for a SB-LOCA counterpart test in reduced (BETHSY and LSTF facilities) and real scales (PWR reactor) (Ciechocki et al., 2022). This case was then also analyzed using the Global Sensitivity Analysis (GSA) methodology. The GSA (Larget and Gautier, 2020) applied by Electricité de France (EDF) and Framatome aimed to identify the most important processes affecting the FoM during a thermal–hydraulic transient. It is focused on the influence of the physical models involved in a well-defined nuclear transient and their impact on the FoM. A recent work applied to Intermediate Break LOCA (IB-LOCA) accidents, proposed by EDF (Geiser et al., 2017; USNRC, 1989) with CATHARE code, aimed to analyze the model's uncertainties and establish a PIRT over different transients.

Work presented in this paper presents results of the GSA applied to the same case previously analyzed by the scaling analysis, a SB-LOCA transient. This latter study was completed to investigate potential complementarities between these two approaches, which are explored and presented hereafter. The rationale for complementing of the scaling with the GSA analysis originates from the inherent nature of both approaches and the observed similarity between the two studies. The scaling analysis is a well-adapted tool to building a well-structured PIRT and analyzing all different phenomena occurring during a transient at a macroscopic scale (via the processes). The sensitivity analysis brings the temporal notion, some kind of parameters interconnection and a quantification of the impact of dominant local processes over a FoM. This paper's novelty lies in its investigation of both the different, common and complementary aspects of the two approaches, aiming to address open questions regarding scaling.

Chapter 2 rapidly introduces the scaling analysis and the sensitivity analysis proposed in this work. The SB-LOCA application case is presented in Chapter 3 where scaling analysis results performed in the past and the new GSA results are presented and analyzed in parallel. This work illustrates the synergy between the two approaches, shares general reflections, and presents guidelines for future applications. Conclusion are presented in Chapter 4.

## 2. Scaling and sensitivity analysis

### 2.1. Scaling analysis

The first step in a scaling analysis consists in the identification of the PoI affecting the Figure of Merit (FoM) and the associated Thermal-Hydraulic Phenomena (THPs). This step is necessary to identify the dominant or the most influent THPs to be respected and consequently to guarantee the good representativeness of an IET to the reactor scale. Some recent works (D'Auria and Bestion, 2022; Bestion, 2024; Bestion et al., 2023) proposed a new method to identify the THPs occurring during the successive PhWs of thermal–hydraulic transients. It establishes a clear link between the PIRT, the choice of PoIs and of scaling equations, and the scaling analysis.

The choice of the scaling equations is important to take into account the FoM and correctly characterize the PoIs for the selected transient. For example, this work presents results for a Small-Break LOCA (SB-LOCA) transient. The FoM is the Peak of Cladding Temperature (PCT) (OECD/NEA, 2017) and the amount of the liquid and the energy transferred in the core are the two important parameters controlling the PCT. As consequence, the mass and energy equations need to be considered. In addition, since the difference between the inlet and outlet mass flow rates in the system is a function of pressure evolution, which affects the activation time of the injection systems (accumulators, high- and low-pressure injection systems), pressure becomes another important parameter to monitor. For this reason, a pressure equation is derived from the energy and mass balance equations and is analyzed in this work.

As presented in [38], the *mass balance* equation within a control volume $V$ for the mixture reads:

$$\dot{M} = \dot{M}_l + \dot{M}_v = \dot{M}_{l,in} - \dot{M}_{l,out} + \dot{M}_{v,in} - \dot{M}_{v,out} \tag{1}$$

where $\dot{M}_k$ is the rate of change of the total fluid mass for the liquid ($l$) and vapor ($v$) phase. Again, starting from the same control volume $V$, the *energy balance* equation may be expressed as follows (energy source due to pumps is here neglected due to the low impact in the selected transient):

$$\frac{dM_l H_l}{dt} + \frac{dM_v H_v}{dt} - V\frac{dP}{dt} = W_w + \dot{M}_{l,in}h_{l,in} - \dot{M}_{l,out}H_l + \dot{M}_{v,in}h_{v,in} - \dot{M}_{v,out}H_v \tag{2}$$

where $H_k$ is the mean specific enthalpy of the phase k. It is supposed that phases exit the volume with the averaged enthalpy of the volume. $W$ is the wall to fluid heating power. Derived from the equation (1) and (2), the average pressure equation becomes:

$$\begin{aligned}(\mu_l M_l + \mu_v M_v)\dot{P} = {} & \dot{M}_{l,in}\left[\nu_l + \nu'_{l,h}\left(h_{l,in} - H_l\right)\right] + \dot{M}_{v,in}\left[\nu_v + \nu'_{v,h}\left(h_{v,in} - H_v\right)\right] \\ & - \dot{M}_{l,out}\nu_l - \dot{M}_{v,out}\nu_v + \nu'_{l,h}W_{wl} + \nu'_{v,h}W_{w,v} + \nu'_{v,h}W_{iv} + \nu'_{l,h}W_{il} \\ & - \frac{W_{iv} + W_{il}}{\varpi} + \frac{W_{w,i}}{\varpi}\end{aligned} \tag{3}$$

With: $\nu'_{k,p} = \frac{\partial v_k}{\partial p}\Big|_{h_k}$ ; $\nu'_{k,h} = \frac{\partial v_k}{\partial h_k}\Big|_{p}$ ; $\mu_k = -\nu'_{k,p} - \nu_k - \nu'_{k,h}$ ; $\overline{\omega} = \frac{h_v - h_l}{v_v - v_l}$

All terms are a volume flow rate or Volume Rate of Change (VRC). Since the volume is constant, the sources of fluid volume have to be compensated by fluid volume sinks. The l.h.s. term is the VRC due to liquid and vapor expansion (or contraction) by pressure change. The r.h.s. terms are respectively the volume flowrate and the associated thermal expansion (or contraction) of entering liquid and vapor, volume flowrate of exiting liquid and vapor, the VRC by thermal expansion (or contraction) of liquid and vapor by wall and interfacial exchange, the VRC by vaporization (or condensation) due to interfacial exchange and the wall boiling/condensation contributions.

Application of advanced methods (H2TS (Jr et al., 2015), FSA (Catton et al., 2009; Novozhilov, 1997), DSS (Reyes, 2015) to the scaling equations using a mature system code allows a classification of the different THPs for a specific thermal–hydraulic transient supporting the establishment of a precise structured Phenomena Identification and Ranking Table (PIRT).

The CEA is working on the development of a scaling tool implemented in the CATHARE code in order to help the user to access to any selected scaling equation terms at system, component or local level and to perform more precise and more reliable scaling analyses.

### 2.2. Statistic tools

The GSA (Larget and Gautier, 2020) is a global statistical approach aimed to identify the impact of different parameters on the FoM. The first step consists in the analysis of the thermal–hydraulic transient and the definition of the FoM. Basing on the expert judgement, a list of input parameters with the corresponding validation ranges are then specified. To that, a PIRT is performed. It allows to define a list of parameters

which has to be evaluated. In this manner, for each accidental transients, a list of impacting parameters and associated models is known. Thanks to the support of mature system codes, a huge number of calculations is performed varying the different parameters in the validation range for the specific thermal–hydraulic transient. At the end of this process, a kind of mesh supplies information on:

- the cross effects between different input variables on the FoM, as function of the temporal phase.
- the monotonous or non-monotonous behavior of an input variable on the FoM and its penalizing zone. The penalizing zone is a region in a model's FoM input parameter space. This zone is characterized by specific input parameters where a particular parameter's influence is heightened and the FoM is maximized. As a result, if the model associated with this parameter is faulty, its impact on the FoM will be more significant.

B. Iooss et al. (Larget and Gautier, 2020) proposed different methods to perform the analysis of the results of the GSA. This work presents both the quantitative and the qualitative approaches in order to obtain a more complete analysis of the studied thermal–hydraulic transient. The qualitative approach based on graphical tools (see Fig. 1: scatter plots and cobweb), allows to evaluate the influent area for each input parameters and their impact to the FoM.

A cobweb or parallel plot is a 2D representation of an n-dimensions space as shown in Fig. 1 on the left. If the vertical axis represents a general parameter (input parameter (Param1, Param2..) or the FoM), the horizontal shows the variation range. An additional information is also the impact of each parameter on the FoM e.g. Param4 is the most influent variable affecting the FoM. The red lines, corresponding to the penalizing zone of the FoM, are totally dispersed for the Param1, Param2 and Param3 over the whole range of variation. Differently, for Param4, they are concentrated in a specific zone corresponding to the penalizing zone of the influent parameters. For Param4, it corresponds to [0.8–1.1]. Since the crossing lines also show the interactions between different parameters, this tool is so useful to rapidly identify the penalizing zone on each variable and the correlation between variables. Fig. 1 on the right shows an example of the pair plot supplying the same information than that previously presented on a different way by aggregation of scatter plot with a variable on the horizontal axis and another variable on the vertical axis. Points in the graph are colored according to the FoM.

The Sobol indexes (Carbiener and Cudnik, 1969) approach was evaluated for this work but it needs a huge number of calculations. The ratio between the information obtained from Sobol indexes compared to the effort in terms of CPU time was low. That is why for this GSA analysis, the Sobol indexes are not used and the graphical tools, the HSIC and PLI indexes, are finally preferred. Concerning the quantitative sensitivity evaluation, three indexes are calculated: the Hilbert-Schmidt Independence Criterion (HSIC – (Da Veiga, 2015; Sueur et al., 2017; Iooss and Lemaitre, xxxx), the p-value of the HSIC index and the Perturbation Law Indices (PLI – (Sueur et al., 2017; Iooss and Lemaitre, xxxx). The HSIC and the PLI are two quantitative evaluations of the dependency between an input parameter and a FoM. The HSIC measures the dependency through non-linear functions. The PLI evaluates the influence of the probability density function characteristics (mean, shape and standard deviation) on the FoM behavior. The p-value evaluated on the HSIC, adds an additional statistical information to the original HSIC criterion. It can be interpreted as the probability of independency between a variable and a FoM based on the HSIC evaluation. In other words, if the p-value is lower than a threshold (empirically equal to 5 %) and so the probability of independency is lower than 5 %, the dependency cannot be rejected. In other words the greater the impact of the model on the FoM, the lower the p-value and the higher the HSIC and PLI index.

By crossing conclusions obtained by this combination of these different tools (graphical tools, HSIC, p-values and PLI), a complete sensitive analysis may be performed.

## 3. Combined application of GSA and the scaling analysis to a SBLOCA

This work intends to demonstrate the synergy between the scaling analysis and the sensitivity analysis for an in-depth understanding of complex thermal–hydraulic transients. Even though scaling analysis is being extensively used in the scientific community and has been the subject of numerous publications in the literature, some scaling issues still need further progress, i.e. i) how to investigate the relationship between the processes and the corresponding closure laws ii) how to quantify the impact and define the acceptability of a distortion for a specific reactor thermal–hydraulic transient iii) what method to assess how processes affect one another and iv) how to determine what controls the timing of the different PhWs. Some reflections are presented hereafter and applied to a concrete case in order to propose some general guidelines in support to the future studies.

### 3.1. SB-LOCA transient for BETHSY, LSTF facilities and a 3 loops-PWR reactor

A. Ciechocki et al. (Ciechocki et al., 2022) performed some scaling analyses for a 6 % cold leg SB-LOCA transient in a PWR based on a counterpart test performed with the LSTF and BETHSY facilities (Ciechocki et al., 2022). Authors applied the FSA method to calculate the dimensionless effect metrics. In a second time, additional calculations are performed by Framatome using the GSA, aimed to confront results, identify synergy and complementarity. In both analyses, simulations are

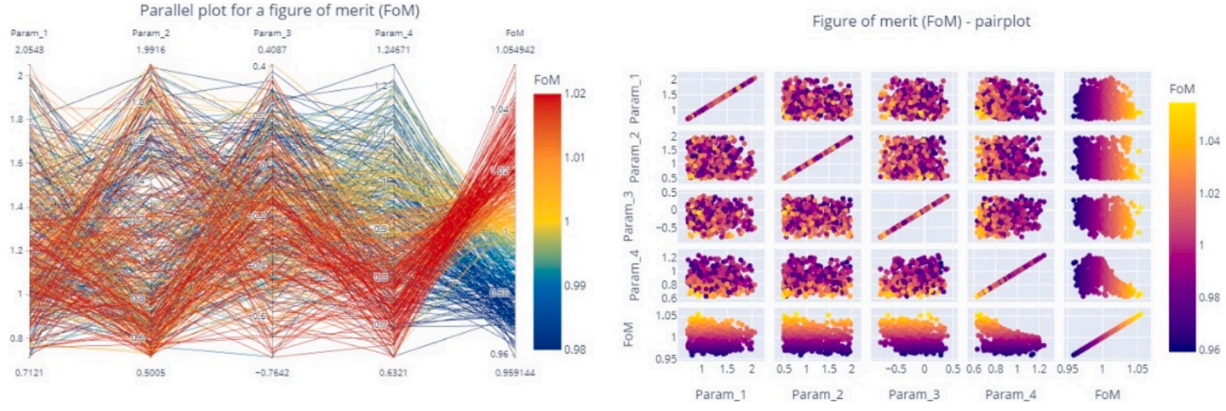



**Fig. 1.** An example to show the qualitative approach of the GSA analysis: cobweb (left) and pair plot (right).

performed with the CATHARE system thermal–hydraulic code.

#### 3.1.1. BETHSY

In the 90′, the French partners Framatome, EDF and CEA performed an important experimental program aimed to understand and validate the French CATHARE code over a wide range of transients in the BETHSY (Boucle d'Etude ThermoHydraulique SYstème) facility. Built in Grenoble, France, BETHSY was designed to simulate a Framatome three loops PWR with a full power of 2775 MWth – 900 MWe. The facility had three loops and one pressurizer and it was a full height and a 1/100 vol scaled at full-pressure reference conditions. The maximal core power was equal to 10 % of the 1/100 nominal power of the reference reactor. i.e. 2.86 MWt. This allows to describe correctly situations with decay power after a scram but induces distortions for transients which start at full power.

#### 3.1.2. LSTF

The LSTF (Large Scale Test Facility) facility was defined by the Japan Atomic Energy Research Institute (JAERI). It represents a Westinghouse four loops PWR with a full power of 3423 MWt – 1100 MWe designed using a power-to-volume scaling factor. It has a full-height scaling and a 1/48 scaled volume at full-pressure reference conditions. This facility has two loops, two steam generators and one pressurizer. The maximum core power is equal to 14 % of the 1/48 nominal power of the reference reactor i.e. 10 MWt.

#### 3.1.3. Thermal-hydraulic transient

A 6 % SB-LOCA transient without High Pressure Safety Injection (HPSI) and Auxiliary Feedwater System (AFW) is analyzed in both facilities as well as in the reference PWR. This transient may be divided in 5 phases:

Phase 1: the Subcooled Blowdown phase (SB). The break is opened. The loss of fluid at the break first empties the pressurizer. Once the pressurizer and its surge line are voided, a two-phase flow establishes in the loops.

Phase 2: the Natural Circulation phase (NC). The pumps being assumed tripped at scram, the flow in the primary loops moves to a natural convection governed by high steam quality and low gravity head between core and SGs and lower quality and higher gravity head between SGs and the core. This phase lasts as long as the primary circuit mass inventory guarantees a two-phase circulation at the top of the steam generator.

Phase 3: Reflux Condenser Mode (RCM). The vapor flow is no more capable to entrain the liquid water to the top of the SG tubes, the natural circulation is stopped. The liquid condensed in the upward part of the steam generator tubes flows to the hot leg in a counter current stream. In parallel, the liquid condensed in the downward part of the SG tubes flows to the intermediate leg and creates a plug. A part of the residual power is released to the steam generators (SGs) and another part is lost at the break. Phase 4: High Quality Discharge at the break (QD). During phase 2 and 3, the terms of the primary pressure equation show some equilibrium between the volume produced in the core (and by heat release from solid structures), and the volume lost by condensation in the SGs and at the break. The primary pressure is almost stabilized in a'pressure plateau' just above the secondary pressure to be able to transfer part of the decay heat to SGs. Once the primary mass inventory sufficiently decreases, the void fraction at the break increases close to the unity and the break volume flowrate increases. The pressure difference between the upper part of the circuit (the lower plenum, the core, the upper plenum, the upper head, the hot legs, the steam generators and the descending part of the intermediate leg) and the lower part (the ascending part of the intermediate legs, the cold legs and the downcomer) increases still to be sufficient to evacuate plugs towards the pressure vessel (Préa et al., 2020). Then the primary pressure decreases below the secondary circuit pressure.

Phase 5: Refilling Phase (RP). The accumulators inject cold liquid water and the primary mass inventory increases. A possible core uncovery with a rod temperature excursion at beginning of Phase 5 is progressively eliminated by core water mass increase.

### 3.2. Comparison between the scaling analysis and the GSA over a concrete case

This paragraph presents the application of the scaling analysis and the GSA on the first three PhWs until the inversion of the primary and the secondary pressures.

In the previous scaling analysis (Ciechocki et al., 2022), the Peak Cladding Temperature (PCT) was defined as the principal FoM for the entire transient and the primary mass inventory and the primary pressure as the main PoIs. A. Ciechocki et al. (Ciechocki et al., 2022) performed a scaling analysis on the entire primary system for the primary mass inventory and primary pressure and on the pressurizer component for the Phase 1. The scaling analysis supported the transient comprehension and contributed to the establishment of a PIRT for the three thermal–hydraulic systems (BETHSY, LSTF and PWR). The comparison between the effect metrics obtained in the IETs and reactor simulations are used to identify the scale distortions.

In the GSA application, the uncertainties are propagated in code simulations for the three scales: BETHSY, LSTF and the 900 MWe reactor. Only parameters having a role on the energy and mass balance equations are taken into account as in the previous work (Ciechocki et al., 2022). The initial steady state conditions are carefully checked for all the simulations so that a modification of a physical closure laws does not satisfy all known initial conditions. For example, uncertainty propagation on heat transfer coefficient in the core can modify the initial temperature of the fuel. It is the same by modifying the heat transfer coefficient in the steam generator. This leads to a deviation on the mean initial primary temperature. In order to ensure the same initial conditions for every calculation, the energy of the primary circuit is preserved for the different calculations. The deviation on the mean temperature can be solved by a secondary pressure adjustment. Five-hundred calculations ran for each scale (BETHSY, LSTF and PWR) on the same scenario. The model uncertainties were established by Inverse Uncertainty Quantification from separated effect test (SET) experimental data compared to calculation results using the CIRCE statistical tool (D'Auria and Galassi, 1998; Ciechocki et al., 2022) or defined by expert judgement. As much as possible, statistical evaluation is applied to reduce the user bias that can arise from expert judgment. The statistical evaluation shows that sufficient experimental data are available to conduct this type of study.

A list of possible processes influencing the FoM and defined by expert judgment is proposed for this work: (1) core heat transfer coefficient (convection, nucleate boiling), (2) steam generator heat transfer coefficient for both primary (liquid convection and the condensation heat transfer) and secondary (liquid convection and the nucleate boiling) circuit. (3) Wall heat transfer coefficient in the whole primary circuit, (4) wall heat transfer specifically in the pressurizer, (5) flashing heat transfer in the vicinity of the break.

#### 3.2.1. Blowdown phase

At the beginning of this phase, the primary loops are still completely liquid except the pressurizer, which controls the primary pressure during this depressurization. As soon as the pressurizer is voided, the 2-phase flow natural circulation starts. Both scaling analysis and sensitivity analyses (GSA) are performed on the pressurizer component. Note that the GSA considers the pressurizer voiding time as the Parameter of Interest for this first phase since it gives an information on the kinetics of the blowdown. Fig. 2 presents the results of this analysis: starting from the higher HSIC and PLI values and the lower p-value, the wall heat transfer in the pressurizer (latent and convective), the flashing at the break and the nucleate boiling in the SG are the most important processes for LSTF. It is the same for BETHSY excepted about the nucleate

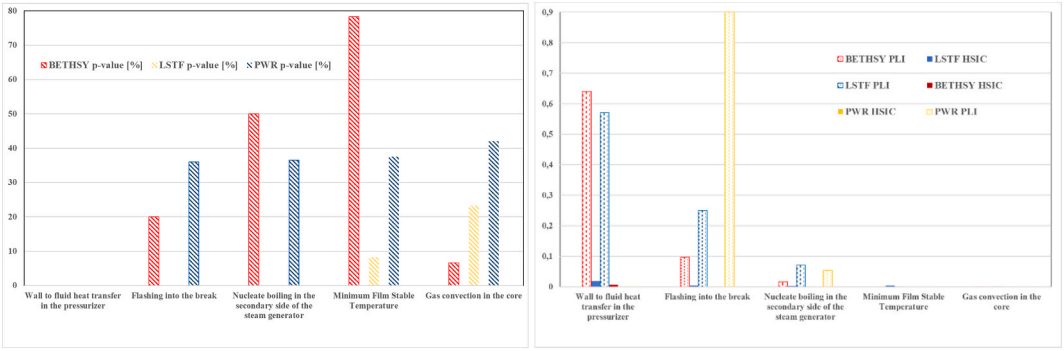



**Fig. 2.** Pressurizer emptying time – Most important parameter for BETHSY (in red), LSTF (in yellow) and PWR (in blue). The HSIC (uniform color), PLI (dotted) and the statistical p-value (striped) are presented. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

boiling in the SG which is not influent. Flashing and nucleate boiling in the SG result as dominant for the PWR.

This result is in agreement with the previous scaling analysis where the volume and the thermal contraction/expansion processes due to interfacial ($Q_{vi}$), ($Q_i$), ($Q_{li}$) and the walls ($Q_{wi}$) heat exchanges in the pressurizer were the most important contributions to the PoI as well as the liquid break flowrate ($Q_l$). Table 1 presents the most important processes (dominant) identified from equation (1) and (3) by the previous scaling analysis. In the PWR as opposed to the IETs, the influence of the vapor volume flowrate from the pressurizer was more significant ($Q_{v_out} = -0.649$). This was attributed (Ciechocki et al., 2022) to the limitations of using the two-node volume module specific to the CATHARE code for the pressurizer because it cannot model the void fraction gradient in the pressurizer and the void entrained in the expansion line may be overestimated. Table 1 also shows some distortions for the vaporization by interfacial transfer that is higher in the PWR (0.584) than in BETHSY (0.448) and LSTF (0.230). The higher contribution of the predicted vapor exiting flow rate for the PWR is associated to the faster depressurization and a more important contribution of flashing. This distortion was in part attributed to the ratio of pressurizer initial mass to the total primary mass that is lower in LSTF (6.38 %) than that in the PWR (8.68 %) and the higher initial secondary pressure fixed as boundary conditions in BETHSY versus LSTF. Since the nominal power is not possible in BETHSY and in LSTF, the choice was made for the counterpart test to respect the integral of the core energy by keeping 10 %NP during a longer time-period than necessary to cross the decay power curve. Moreover, the pump initially produces only 10 % of nominal flow rate in experiments and the speed rapidly decreases to zero. The scaling analyses and the *generalized distortion correction strategy* under development at CEA showed (Ciechocki et al., 2022) that the choice of the power and the modeling of the pump created a significant distortion between the processes occurring in the IETs and the PWR. The distortion is so corrected in the simulation, by changing the input deck and its impact on the entire transient is studied. A comparison between the distorted and the corrected transient is investigated. This approach makes it possible to quantify the distortion's effect on the transient and better assess its overall importance.

Despite the consistency between results obtained from the GSA and the scaling analysis, one can note that scaling analysis permits to take into account all THPs occurring within the PhW that is not so trivial with the GSA since the list of phenomena is a-priori established by expert judgement.

Concerning the GSA, Fig. 3 shows the pressurizer voiding time in function of the wall to fluid heat transfer in the pressurizer for BETHSY, LSTF and the reactor case. Since the impact of parameters over the FoM is indicated by the very dense cloud of results along a certain trend, Fig. 3 is consistent with Table 1 showing that the heat transfers (convective and latent) in the pressurizer is an important parameter. Figures also show that a higher wall heat transfer reduces the pressurizer blowdown duration. The growth of the heat transfer coefficient induces a higher vapor production in the pressurizer and this effect produces a quicker voiding of this component. The dependency is rather linear for the IETs but not for the PWR where the increase of the heat transfer has no more influence for values lower than the unity. An independent zone for the heat exchange in the pressurizer is reached above a certain limit. Unfortunately, no further studies were conducted to investigate this aspect.

Concerning the flashing at the break, the increase of the flashing flux produces more vapor and reduces the critical flow at the break. Fig. 4 shows this dependency and the importance of this parameter in relation to the FoM.

### 3.2.2. *From natural circulation phase to the end of the pressure equilibrium between the primary and the secondary system*

Once the pressurizer is completely empty and pumps stop, the Natural Circulation (NC) starts. The primary circulation is so guaranteed by the different density between the hot and the cold part of circuit. Vapor slowly appears at the break. As long as the fluid volume production by the core boiling and by other wall heating is not compensated by the fluid volume lost at the break, the primary pressure remains above the secondary pressure to allow a fluid volume sink by transfer energy to the SG. The liquid mass decreases, the void fraction in the loops increases

**Table 1**
Important processes identified during à 6 % SB-LOCA in BETHSY, LSTF and the reactor PWR. Dominant (1-DPh), second, third and fourth important phenomena (2-DPh, 3-DPh, 4-DPh) during the blowdown phase. PoI = M (masse): contribution to the living vapor or liquid (k-phase) flowrate $\dot{M}_k$ . PoI=P (pressure): contribution to the living vapor or liquid (k-phase) flowrate $\dot{Q}_k$ , thermal expansion/contraction interphase/vapor (or liquid) heat transfer $Q_{ki}$, volume change by interfacial heat exchange at the free surface $Q_i$, and volume change by wall boiling or condensation $Q_{wi}$.

| | PoI: M1 | | PoI: $P_{PRZ}$ | | | | |
|---|---|---|---|---|---|---|---|
| | 1-DPh | 2-DPh | 1-DPh | 2-DPh | 3-DPh | 4-DPh | 5-DPh |
| **BETHSY** | $\dot{M}_{l_out}(-1.000)$ | $\dot{M}_{v_out}(-0.002)$ | $Q_{l_out}(-1.000)$ | $Q_i(0.448)$ | $Q_{v_out}(-0.298)$ | $Q_{vi}(0.290)$ | $Q_{wi}(0.163)$ |
| **LSTF** | $\dot{M}_{l_out}(-1.000)$ | $\dot{M}_{v_out}(-0.001)$ | $Q_{l_out}(-1.000)$ | $Q_{vi}(0.466)$ | $Q_{v_out}(-0.269)$ | $Q_i(0.230)$ | $Q_{li}(-0.067)$ |
| **PWR** | $\dot{M}_{l_out}(-1.000)$ | $\dot{M}_{v_out}(-0.004)$ | $Q_{l_out}(-1.000)$ | $Q_{v_out}(-0.649)$ | $Q_i(0.584)$ | $Q_{vi}(0.437)$ | $Q_{li}(-0.102)$ |

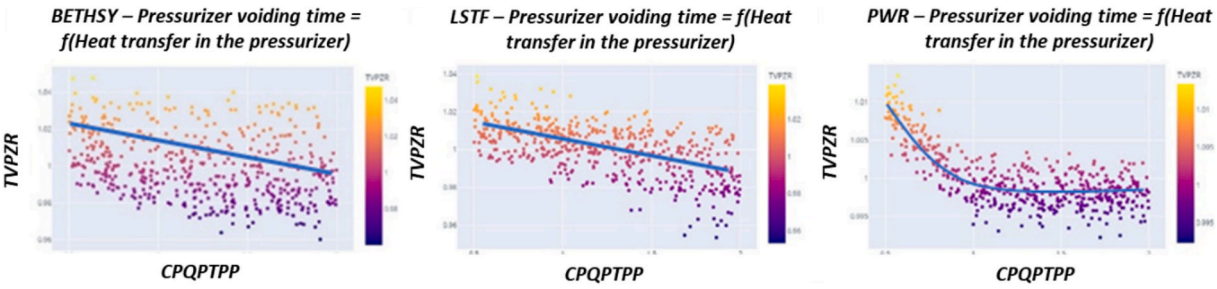



**Fig. 3.** Pressurizer voiding time (TVPRZ) in function of the wall to fluid heat transfer (CPQPTPP) in the pressurizer for BETHSY, LSTF and 900 Mwe PWR.

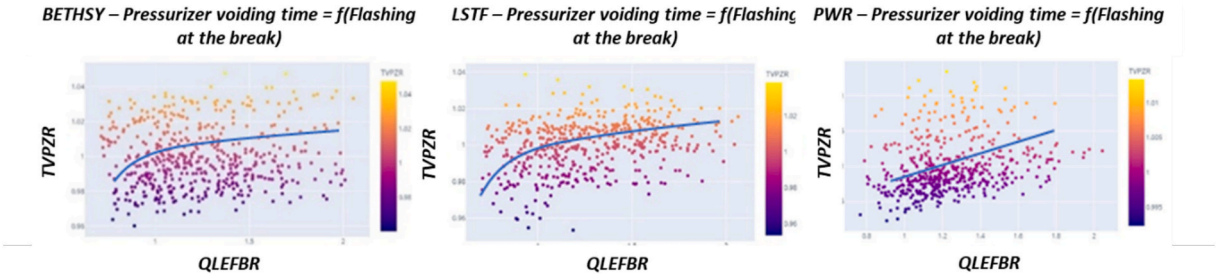



**Fig. 4.** Pressurizer voiding time (TVPRZ) in function of the flashing in the pressurizer (QLEFBR) for BETHSY, LSTF and 900 Mwe PWR.

**Table 2**
Important processes identified during à 6 % SB-LOCA in BETHSY, LSTF and the reactor PWR. Dominant (1-DPh), second and third important phenomena (2-DPh and 3-DPh) during the reflux condenser phase. PoI = M (masse): contribution to the living vapor or liquid (k-phase) flowrate $\dot{M}_k$ . PoI=P (pressure): contribution to the living vapor or liquid (k-phase) flowrate $\dot{Q}_k$ , volume change by interfacial heat exchange $Q_i$, thermal expansion/contraction by wall-vapor heat transfer $Q_{wk}$ and volume change by boiling or condensation $Q_{wi}$.

| | PoI: M1 | | PoI: P1 | | | | |
|---|---|---|---|---|---|---|---|
| | 1-DPh | 2-DPh | 1-DPh | 2-DPh | 3-DPh | 4-DPh | 5-DPh |
| BETHSY | $\dot{M}_{l_out}(-1.000)$ | $\dot{M}_{v_out}(-0.218)$ | $Q_{v,break}(-1.000)$ | $Q_{wi,core}(0.997)$ | $Q_{i,SG}(-0.624)$ | $Q_{l,break}(-0.245)$ | $Q_{wv,core}(0.138)$ |
| LSTF | $\dot{M}_{l_{out}}(-1.000)$ | $\dot{M}_{v_out}(-0.293)$ | $Q_{v,break}(-1.000)$ | $Q_{wi,core}(0.796)$ | $Q_{i,SG}(-0.634)$ | $Q_{l,break}(-0.193)$ | $Q_{i,sys}(0.115)$ |
| PWR | $\dot{M}_{l_out}(-1.000)$ | $\dot{M}_{v_out}(-0.293)$ | $Q_{v,break}(-1.000)$ | $Q_{wi,core}(0.727)$ | $Q_{i,SG}(-0.285)$ | $Q_{l,break}(-0.229)$ | $Q_{i,sys}(0.065)$ |

until vapor is no more capable to entrain liquid water to the top of SG. The natural circulation finishes and the Reflux Condenser Mode (RCM) starts. During this phase, the condensate liquid flows down the ascending part of the U-tubes, returning to the HL, while the liquid flowing in the descending part accumulates in the Intermediate Leg (IL). For this CL-SBLOCA, the Loop Seal Plugging (LSP) separates the HL and CL with a consequent pressure reduction in the CL due to the loss of fluid volume at the break. At the same time, pressure in HL stabilizes when the fluid volume change by boiling is compensated by condensation in the SGs plus some losses of volume through the bypass from upper head to downcomer. Pressure difference between CL and HL increases until it pushes the liquid water in the IL to the CL and to the RPV. This is called Loop Seal Clearing (LSC). The timing of this phenomenon affects the dry-out duration and the PCT. An important bifurcating event occurs after this LSC: the increased void fraction at the break leads to greater pressure loss and a primary/secondary pressure inversion. Results of the scaling analysis (Ciechocki et al., 2022) shows that the dominant phenomena during the RCM over the primary depressurization are the vapor flow at the break, the volume change by wall boiling in the core and the volume change by interfacial heat transfer in the SG. The liquid flow at the break is also an important phenomenon for the three systems.

Table 2 shows the dominant processes for the reflux condensation phase obtained during the previous work (Ciechocki et al., 2022). Table shows a first distortion on the volume change by wall boiling in the core ($Q_{w,i,core}$) between the IETs (LSTF: 0.796, BETHSY: 0.997) and the PWR (0.727). In particular, this term is higher in BETHSY than that in LSTF and the PWR. The thermal expansion by wall-vapor in the core is influent on BETHSY (0.138) but considered negligible for the LSTF and the PWR. This discrepancy may be related to a longer core dry out period observed in BETHSY facility, a super-heated vapor production and a delay on the pressure inversion time.

For the GSA, the pressure inversion time is chosen as the FoM for this specific PhW. Fig. 5 shows the quantitative indexes of the GSA describing the relative influence of the parameters on the FoM. The most important phenomena are selected starting from a higher HSIC and PLI coefficient and a lower p-value.

As shown in Fig. 5, the flashing at the break and the nucleate boiling in the SG secondary side are the most influent parameters on LSTF and the PWR reactor. On the other side, for BETHSY, the most important parameters are the flashing at the break and the minimal stable film temperature (MSFT) in the core, in agreement with the previous scaling analysis.

Fig. 6 shows the impact of the flashing and the MSFT over the primary/secondary pressure inversion time for BETHSY facility. The inversion time is proportional to the flashing at the break and the MSFT.

On the contrary, the MSFT parameter is not important for LSTF and the PWR. Fig. 7 and Fig. 8 show the impact of the flashing and the nucleate boiling models over the pressure inversion time for LSTF and PWR. Despite a slightly dispersion of results, a main trend is observed; the increase of the nucleate boiling at the secondary side of the steam generator anticipates the primary pressure inversion.

It was observed that pumps in the IETs have a very low inertia in

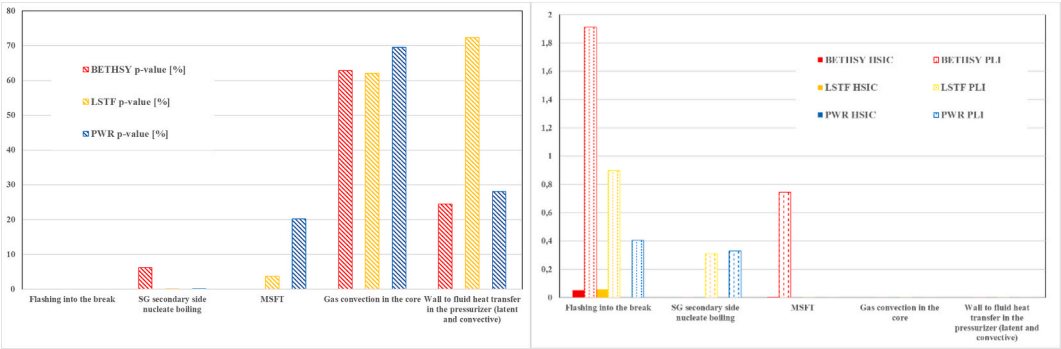



**Fig. 5.** Primary/secondary pressure inversion time – Most important parameter for BETHSY (in red), LSTF (in yellow) and PWR (in blue). The HSIC (uniform color), PLI (dotted) and the statistical p-value (striped) are presented. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

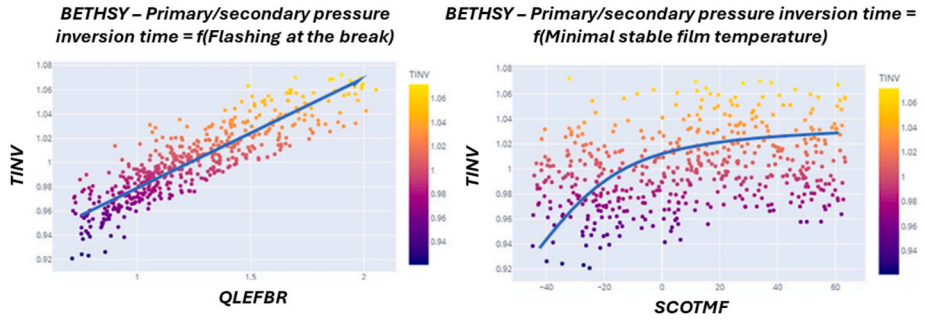



**Fig. 6.** Primary/secondary pressure inversion time as a function of flashing at the break (left) and minimal stable film temperature in the core (right) for BETHSY.

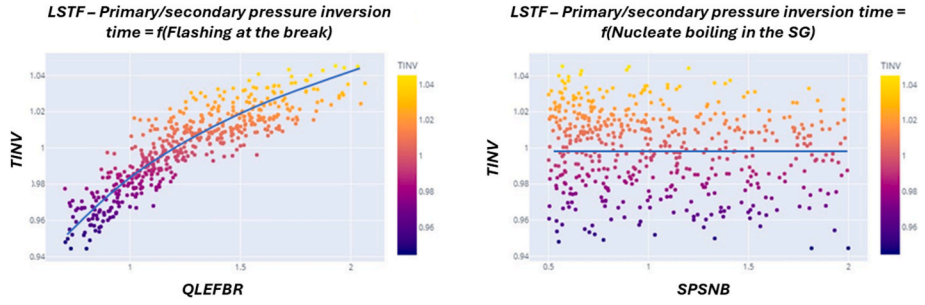



**Fig. 7.** Primary/secondary pressure inversion time as a function of flashing at the break (left) and steam generator nucleate boiling (right) for LSTF.

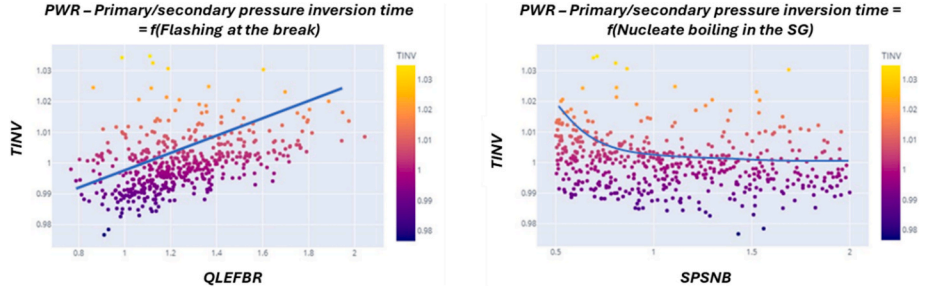



**Fig. 8.** Primary/secondary pressure inversion time as a function of flashing at the break (left) and steam generator nucleate boiling (right) for PWR.

comparison to the PWR case (Ciechocki et al., 2022). As consequent, there is no initiation of water plugs in the U-leg of the PWR and the dry-out duration is short. For BETHSY and LSTF, water plugs and the LSC time strongly affects the PCT and the primary/secondary pressure inversion time. If a core uncovering is observed before the LSC, as for BETHSY, and for a long period, the cladding rewetting produces over heated vapor that maintains higher the primary pressure and postpones the pressure inversion with a consequent higher PCT. It explains the influence of the MSFT on Fig. 6. By increasing the MSFT value, the rewetting is obtained at higher temperatures. It anticipates the rewetting and increases the amount of vapor produced, with a consequent increase of a higher primary pressure and a delayed inversion. For low MSFT,

since the cladding cooling is longer and the rewet is delayed, the pressure inversion is obtained before the rewetting and the primary pressure is not highly maintained.

### 3.3. Some guidelines for the application of the scaling and the sensitivity analysis

Fig. 9 illustrates, in a schematic way, synergies between the scaling (in blue) and the GSA (in grey) and it shows how the GSA may support the scaling analysis. The previous application to a SB-LOCA points out the contribution of both approaches and suggests some guidelines in order to better perform a scaling analysis. Nevertheless, this concrete case shows as some additional efforts are necessary to better integrate the GSA and the scaling analysis for the future applications. Both approaches should be conducted in parallel to enhance coordination and improve the understanding and completeness of the results.

As shown in Fig. 9, the first steps of a scaling analysis (Zuber et al., 2007) for a specified transient consists in the definition of the FoM, the PoIs and the scaling equations in order to correctly take into account all processes involved. The thermal–hydraulic transient is divided in some (e.g. 3, 4 or 6) time phases, also called PhWs, governed by a few influent THPs. Each PhW is delimited by some important Bifurcating Events (BEs) that strongly change the evolution of the transient. Several scaling methods exists in such a way as to find dimensionless numbers characterizing the specific process $k$ occurring during the transient (e.g. the Effect Metric '$\Omega_k$' for the FSA or the Pi-Groups '$\Pi_k$' for the H2TS). Thanks to these dimensionless numbers, the scaling analysis may contribute to i) better understand the key influences in a complex thermal–hydraulic transient, (Step 5a) ii) establish a PIRT (Step 6a) and iii) support the facility design or iv) investigate scaling distortions (Step7a) in a-posteriori approach.

The GSA may contribute to this analysis by adding complementary information in order to support the understanding of the thermal–hydraulic transient (Step 5a). For example, the GSA completes the understanding of the scaling processes and the possible corresponding distortions among different scaled systems, supplying supplementary information on corresponding closure laws and easily shows connections between the different models and the FoM. An example is shown in Fig. 1 with the Cobweb graph. Although in the GSA, the list of potentially important processes is provided at the beginning of the analysis based on expert judgment and is not a result of the study as in the scaling analysis, the final PIRT (Step 6b) is consistent with the scaling analysis. The penalizing zone of a model corresponds to the range where the model can vary, leading to a consequential impact on the FoM. Definition of this amount is important from a safety point of view. The GSA immediately shows this zone and identifies the variation ranges. Once the distortions are identified by the scaling analysis (Step 7a), the GSA allows identifying the influential range of each model and it helps to quantify the importance of each distortion (Step 7b). The *generalized distortion correction strategy* proposed by CEA consists in using hybrid calculations where the initial conditions or the system design (Zuber et al., 2007) or again a closure law (with the GSA) may be modified by changes in the input deck. Results of these hybrid calculations reveal the impact of scaling distortions on the FoM and PoIs for a specific transient. The GSA permits to identify models that highly influence the FoM and highlights the possible distortions as shown in Fig. 3 where the different trend for the PWR suggests a different dependency between the studied closure law and the FoM.

The scaling analysis is performed first at system scale to identify the dominant processes and quantify the distortions. Using mature system codes for estimating the process terms of scaling equations is more precise than a simple dimensional analysis and an average estimation of terms during a transient phase. The mature code can predict the time evolution of each term and this helps understanding what controls some bifurcating events such as the end of the first depressurization, the duration between LSP and LSC, the crossing time of primary and secondary pressures. Moreover, a systematic use of hybrid calculations that correct, in IET calculations, the distortions one by one will provide a clear understanding of what explains the differences between IET results and the code simulations of the reactor transient. In addition, these hybrid calculations will provide new more efficient acceptability criteria for scaling distortions since they show the effect of each distortion on the Figure of Merit itself. This still requires some other tools to be implemented in the CATHARE system code to further extending the current capabilities to correct the distortions in a simple way for the code user. The scaling analysis should in principle be performed from system scale to component and local scales in the top-down approach but it appears that the GSA can give an easier access to the impact of local processes

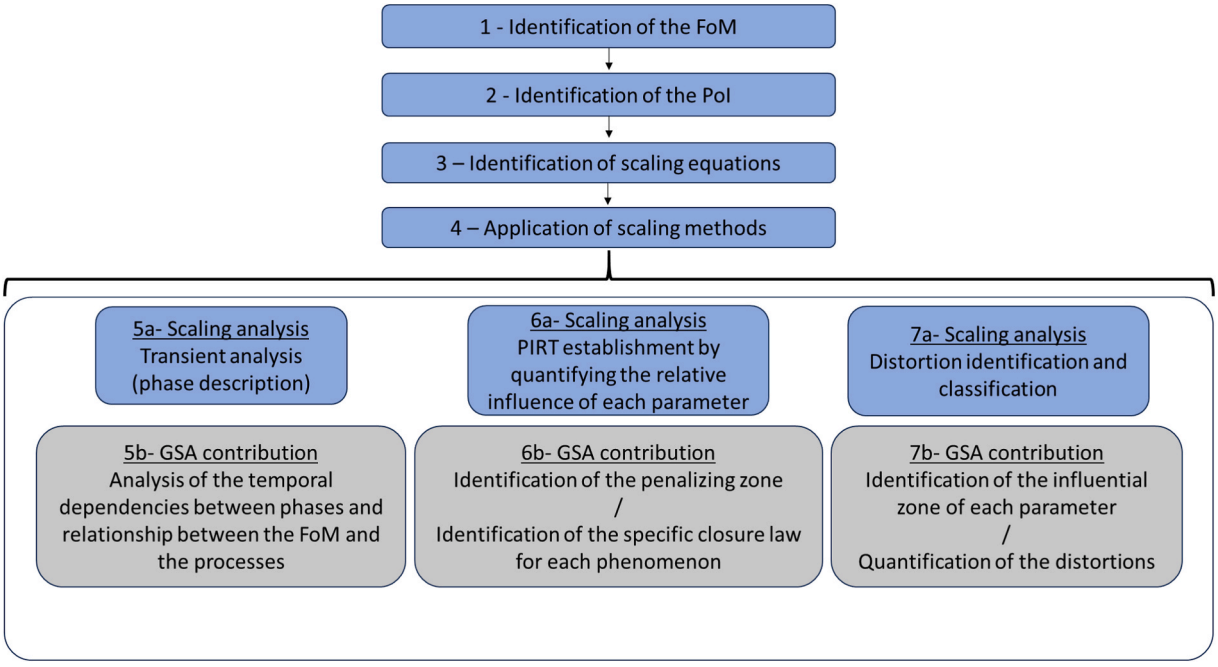



**Fig. 9.** Diagram of Scaling and GSA contributions.

modeled in code closure laws on the important transient parameters and on the Figure of Merit.

The complementarity of the two approaches provides an in-depth identification of the governing phenomena at every scale from system to local. It contributes to explain the differences between an IET and a reactor simulation and helps defining priorities for code validation and model scalability demonstrations.

## 4. Conclusions

This paper shows the synergies and the complementarity of using both a scaling analysis performed with a mature system code and a Global Sensitivity Analysis (GSA) to support the simulation of a reactor transient. Based on previous scaling analysis results for a 6 % cold leg break LOCA on a PWR, BESTHY and LSTF facilities, a new application of the GSA methodology to the same case is presented and investigated. By analyzing the same transient with both approaches, it was possible to begin reflecting on their potential complementarity. Some reflections and guidelines are so presented in this paper for the future works in order to improve the scaling analysis in support to the BEPU methodology.

The scaling analysis is the best tool to identify dominant phenomena at the macroscopic scales during a thermal–hydraulic transient; the GSA complements the understanding of the transient, supplying additional information on the dominant processes and their interactions at the most local scale. More precisely the scaling analysis contributes to the understanding of all processes involved in a thermal–hydraulic transient and supports the establishment of a structured PIRT starting from appropriated scaling equations. During a specific PhW in one or more scaled facilities, dominant physical processes are identified and quantified. This approach also helps to pinpoint possible distortions and thanks to a hybrid calculation such as the *generalized distortion correction strategy* under development at CEA, it provides a better understanding of the transient and the distortion quantification. This paper first shows that applying both the scaling and the GSA leads to the same PIRT and suggests similar possible distortions even though with some differences. If the scaling gives a more general information about processes (a kind of top-down approach) the GSA provides a more precise quantification of the parameters' influence on a specific Figure of Merit (FoM) via closure laws (bottom-up). If the scaling processes are derived from specific scaling equations, a pre-defined list based on expert judgment is supplied at the beginning of the GSA analysis. Some works are ongoing at CEA aimed to delve deeper into the distortion quantification.

Despite its interesting potential, this study requires further investigation. Additional transients should be investigated, with both approaches being performed simultaneously. In addition, the GSA seems to be a very interesting tool to support the quantification of scaling distortions. It gives access to key information—such as closer laws, sensitive models, and penalizing zones—that is important for identifying the importance of a specific distortion. Lastly additional efforts are also necessary to better investigate the temporal and the inter-processes dependency contribution of the GSA among the different PhW.

## CRediT authorship contribution statement

**S. Carnevali:** Writing – original draft, Supervision, Methodology, Investigation, Formal analysis. **P. Gaillard:** Writing – review & editing, Resources, Investigation, Formal analysis. **J.L. Vacher:** Writing – review & editing, Resources, Investigation, Formal analysis. **T. Fercoq:** Investigation. **D. Bestion:** Writing – review & editing, Methodology, Formal analysis.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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## Data availability

The data that has been used is confidential.

## References

Bestion, D., 2024. On some was to improve BEPU methodologies. Part 2: Guidelines on V&V&UQ and code scalability demonstration. BEPU, Lucca.
Bestion, D., D'Auria, F., 2022. Identification of Thermal-Hydraulic Phenomena for PWR transient analysis and simulation. NURETH-19. Brussels, Belgium, March 6–11.
Bestion, D., D'Auria, F., Carnevali, S., Ciechocki, A., 2024. On some was to improve BEPU methodologies. Part 1: Guidelines on PIRT –Transient analysis – scaling analysis - IET and SET design. BEPU, Lucca.
Carbiener, W.A., Cudnik, R.A., 1969. Similitude considerations for modeling nuclear reactor blowdown. Trans. Am. Nucl. Soc. 12, 361.
Carnevali, S., 2024. Some reflections and developments for phenomena identification and scaling analysis applied to thermal-hydraulics of nuclear reactors. ATH-2024, Orlando, Floride.
Catton, I., Zuber, N., Wulff, W., Rohatgi, U.S., 2009. Application of fractional scaling analysis (FSA) to loss of coolant accidents (LOCA); component level scaling for peak clad temperature. J. Fluids Engineering.
Ciechocki, A., Carnevali, S., Bestion, D., Rossi, L., 2022. Scaling analysis of the loop seal plugging and clearing phenomena in a SB-LOCA transient on the BETHSY facility. Nucl. Eng. Design 400.
Ciechocki, A., Carnevali, S., Bestion, D., Rossi, L., 2022. Scaling analyses of a SB-LOCA counterpart test between BETHSY and LSTF facilities and a three loops PWR. NED 400.
Ciechocki, A., Carnevali, S., Bestion, D., Rossi, L., 2024. Scaling analyses of a SB-LOCA counterpart test between BETHSY and LSTF and a three loop PWR. Nucl. Eng. Design 422.
D'Auria, F., Bestion, D., 2022. Nuclear thermal-hydraulic phenomena: bases and challenges. Nucl. Technol.
D'Auria, F., Galassi, G.M., 1998. Code validation and uncertainties in system thermal hydraulics, progress. Nucl. Energy 33 (1/2), 175–216.
D'Auria, F., Camargo, C., Mazzantini, O., 2012. The best Estimate Plus uncertainty (BEPU) approach in licensing of current nuclear reactors. J. Nucl. Eng. Design 248.
Da Veiga, S., 2015. Global sensitivity analysis with dependence measures. J. Stat. Comput. Simul. 85 (No. 7), 1283–1305.
Geiser, H., Vacher, J.-L., Robiolo, P., 2017. The use of integral effects tests for the justification of new evaluation models based on the BEPU approach. NURETH-17, September 2017, Xi'an, China.
Gretton, A., Bousquet, O., Smola, A., Schölkopf, B., 2005. Measuring statistical dependence with Hilbert-Schmidt norms. In: Proceedings Algorithmic Learning Theory. Springer-Verlag, pp. 63–77.
Bestion, D., Carnevali, S., Ciechocki, A., 2023. The use of System Codes for Scaling Analysis and the use of Scaling tools for the Analysis of Code Predictions. NURETH-20, Aug 2023, Washington, United States.
B. Iooss and P. Lemaitre, A review on global sensitivity analysis methods, HAL open science, https://hal.science/hal-00975701.
IRSN : Guide de l'ASN n°28 : Qualification des outils de calcul scientifique utilisés dans la démonstration de sûreté nucléaire, 2017.
Ishii, M., Revankar, S.T., Leonardi, T., Dowlati, R., Bertodano, M.L., Babelli, I., Wang, W., Pokharna, H., Ransom, V.H., Viskanta, R., Han, J.T., 1998. The three-level scaling approach with application to the Purdue University Multi-Dimensional Integral Test Assembly (PUMA). Nucl. Eng. Des. 186, 177–211.
Joon Yoon, H., Alyammahi, N., Al-Yahia, O., Leung, R., 2022. The scaling effect analysis of counterpart test between LSTF and ATLAS IET facilities on 1% top head break LOCA. NED 397.
Jr, R.J.N., Frepoli, C., Yurko, J., 2015. The dynamical system scaling methodology: comparing dimensionless governing equations with the H2TS and FSA methodologies. International Topical Meeting on Nuclear Reactor Thermal Hydraulics (NURETH-16), Chicago, (Ill, US).
V. Larget1, M. Gautier, Increasing conservatism in BEPU IB LOCA safety studies using complementary and industrially cost effective statistical tools, BEPU 2020 Giardini Naxos, Sicily, Italy, May 17-22, 2020.
Liu, T.-J., Lee, C.-H., Way, Y.-S., 1997. IIST and LSTF counterpart test on PWR station blackout Transient. J. Nucl. Eng. Design 167.
Liu, T.J., Lee, C.H., Chang, C.Y., 1998. Power-operated relief valve stuck-open accident and recovery scenarios in the Institute of Nuclear Energy Research integral system test facility. J. Nucl. Eng. Design 186.

Martinez-Quiroga, F., Reventos, J., 2014. Freixa, applying UPC scaling-up methodology to the LSTF-PKL counterpart test. Sci. Technol. Nucl. Install. 1, 1–18.
Nahavandi, A.N., Castellana, S., Moradkhaniav, E.N., 1979. Scaling laws for modeling nuclear reactor systems. J. Nucl. Sci. Eng. 72.
Novozhilov, I., 1997. Fractional Analysis – Methods of Motion Decomposition. Birkhauser Publisher, Boston (Ma US).
OECD/NEA, 2017. A state-of-the-art report on scaling in system thermal-hydraulics applications to nuclear reactor safety and design. Nuclear Safety, NEA/CSNI/R (2016)14.
Préa, R., Fillion, P., Matteo, L., Mauger, G., Mekkas, A., 2020. CATHARE-3 V2.1: the new industrial version of the CATHARE code. In: Proceedings of Advances in Thermal Hydraulics (ATH'20), pp. 730–742.
Reyes, J.N., 2015. The dynamical system scaling methodology. NURETH-16.
Reyes, J.N., Frepoli, C., 2015. The dynamical system scaling methodology compararing dimensionless governing equations with the H2TS and FSA methodologies. In: NURETH-16, p. 15.
Sueur, R., Iooss, B., Delage, T., 2017. Sensitivity Analysis using Perturbed-Law based Indices for quantiles and application to an industrial case. Proceedings of the 10th International Conference on Mathematical Methods in Reliability (MMR 2017).
U.S. NRC, 2005. Regulatory guide 1.203, Transient and accident analysis methods. URL, https://www.nrc.gov/docs/ML0535/ML053500170.pdf.
USNRC, 1989. Quantifying Reactor Safety Margins: Application of CSAU to a LBLOCA. NUREG/CR-5249 Washington, USA.
Zuber, N., 1991. Appendix D: a hierarchical, two-tiered scaling analysis, an integrated structure and scaling methodology for severe accident technical issue resolution. US NRC, NUREG/CR-5809 (1991). USNRC, Washington (D.C US).
Zuber, N., Rohatgi, U.S., Wulff, W., Catton, I., 2007. Application of fractional scaling analysis (FSA) to loss of coolant accident (LOCA): methodology development. J. Nucl. Eng. Design 237.